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310401

Roll No. _____

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B. Tech. III Sem. (Main) Exam., Dec. - 2019

Common for CS/IT

3CS2-01 Advance Engineering Mathematics

Time: 3 Hours

Maximum Marks: 120

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 ×2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions 5 ×8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 ×15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

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PART - A

Q.1 Define random variable.

Q.2 What is the coefficient of skewness, if the mean and mode of the distribution are equal? 0

Q.3 If mean of Poisson distribution is 3, then what is the value of variance? 3

Q.4 Define the uniform distribution. $\frac{1}{b-a}$

Q.5 What is optimization?

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Q.6 Write at least two methods for solving a multivariable optimization problem with equality constraints.

Q.7 Consider the following problem –

Minimize $z = f(X)$,

Subject to $g_j(X) \leq 0$; $j = 1, 2, 3, \ldots, m$.

Then write the suitable Kuhn – Tucker conditions.

Q.8 What is difference between a slack and surplus variable?

Q.9 If the given LPP has an optimal solution, then what about the solution of dual problem?

Q.10 Define unbalanced transportation problem.

PART – B

Q.1 The joint probability density function of a two-dimensional random variable (X, Y) is given by http://www.mgsuonline.com

$$f(x,y) = \begin{cases} 2, & 0 < x < 1,\ 0 < y < x \\ 0, & \text{elsewhere} \end{cases}$$

Find the marginal density functions of X and Y. Also find the conditional density function of Y given $X = x$ and conditional density function of X.

Q.2 Derive moment generating function for Binomial distribution. Hence, find mean and variance for the same.

Q.3 Suppose on an average 1 house in 1000 in a certain district has a fire during a year. If there are 2000 houses in that district, what is the probability that exactly 5 houses will have a fire during the year?

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Q.4 A firm manufactures 3 products A, B and C. The profits are ₹ 3, ₹ 2 and ₹ 4 respectively. The firm has 2 machines and below is the required processing time in minutes for each machine on each product.

		Product		
		A	B	C
Machine	G	4	3	5
	H	2	2	4

Machine G and H have 2,000 and 2,500 machine-minutes respectively. The firm must manufacture 100 A's, 200 B's and 50 C's, but no more than 150A's. Formulate the mathematical model to maximize profit.

Q.5 A beam of length l is supported at one end. If v is the uniformly distributed load per unit length and the bending moment M at a distance x from the end is given by

$$M = lx - \frac{1}{2}\omega x^2,$$

then find the maximum bending moment.

Q.6 Using the Lagrange's multiplier method, find the minimum value of $x^2 + y^2 + z^2$ when $ax + by + cz = p$.

Q.7 Using Simplex method, show that the following linear programming problem has an unbounded solution:

Maximize $z = x_1 + 2x_2$

Subject to $x_1 - x_2 \leq 10$

$3x_1 - 2x_2 \leq 40$

And $x_1, x_2 \geq 0$

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PART – C

Q.1 The first four moments of a distribution about the point 4 of the variable are – 1.5, 17, - 30 and 108. Find the moments about the mean, β_1, β_2 and comment on their values. Also find the moments about the point 2.

Q.2 Calculate the coefficient of correlation from the following data:

X:	1	2	3	4	5	6	7	8	9
Y:	9	8	10	12	11	13	14	16	15

Also obtain the equations of line of regression and obtain an estimate of Y which should correspond on the average to X = 6.2.

Q.3 Write a short note on the classification of optimization problems based on various parameters.

Q.4 Use Kuhn – Tucker conditions to minimize $f(x, y) = 10x - x^2 + 14y - y^2$, where $x \leq 10$, $x + y \leq 8$ and $x, y > 0$. http://www.mgsuonline.com

Q.5 (a) Determine an initial basic feasible solution of the following problem by the method of North – West Corner Rule:

	D_1	D_2	D_3	D_4	a_i
O_1	4	6	9	5	16
O_2	2	6	4	1	12
O_3	5	7	2	9	15
b_j	12	14	9	8	43

(b) Write the dual of the following problem:

Minimize $z = 2x_1 + x_2$

Subject to $3x_1 + x_2 \geq 3$, $4x_1 + 3x_2 \geq 6$, $x_1 + 2x_2 \geq 2$

and $x_1, x_2, x_3 \geq 0$

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